

Lecture Three (153-248)

(Sept 03)

Linear Regression Math

Simple Question 1: Suppose I have two numbers a & b . I tell you that $a + b = 7$. What can you say about a ?

Simple Question 2: I have two numbers a & b . Suppose b is small in magnitude. As before, $a + b = 7$. What can you say about a ?

Solution: a & b .

Assume $a \sim \text{Unif}(-C, C)$ ✓
independent for a very large C .

$$b \sim N(0, \sigma^2) \quad \sigma^2 = 0.5$$

Calculate $a \mid a + b = 7$

Let $y = a + b$.

$$a \mid y = 7$$

$$y = \underset{\text{Unif}(-C, C)}{\uparrow} a + \underset{N(0, \sigma^2)}{\downarrow} b$$

$$f(a) \\ a|y=7$$

$$y|a$$

$$= \frac{f(7|a) f(a)}{f(7)}$$

$$f_y(7)$$

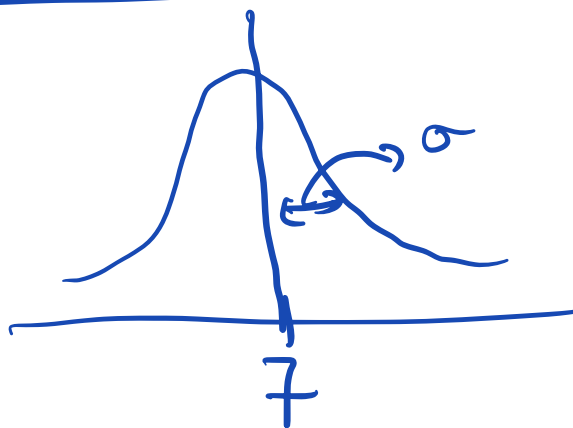
$$\propto f_y(7) f_a(a)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(a-y)^2}{2\sigma^2}\right) \frac{I(-c < a < c)}{2c}$$

$$\propto \frac{1}{\sigma} \exp\left(-\frac{(a-7)^2}{2\sigma^2}\right) \quad \hat{y}=7$$

$$a|a+b=7 \sim N(7, \sigma^2)$$

Answer: $N(7, \sigma^2)$



$$y|a \sim N(a, \sigma^2)$$

Bayes Rule

Question 3: I have a, b_1, b_2 .
 b_1 & b_2 are small in magnitude.

$$a + b_1 = 7 \checkmark$$

$$a + b_2 = 10 \checkmark$$

What can you say about a ?

Solution: Assume

independent

$$a \sim \text{Unif}(-C, C)$$

$$b_1, b_2 \sim N(0, \sigma^2)$$

$$\log \sigma \sim \text{Unif}(-C, C)$$

a

$$a + b_1 = 7$$

$$a + b_2 = 10$$

$$y_1 = \underline{a} + b_1$$

$$y_2 = a + b_2$$

$$a \mid y_1, y_2$$

$$a, \sigma \mid y_1, y_2$$

$$f(a, \sigma) \propto f_{y_1, y_2 \mid \underline{a}, \underline{\sigma}}(y_1, y_2) \underbrace{f(a, \sigma)}_{\underline{a}, \underline{\sigma}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_1 - a)^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_2 - a)^2}{2\sigma^2}\right)$$

$$\frac{I(-C < a < C)}{2C} \frac{I(-C < \log \sigma < C)}{2C} \times \frac{1}{\sigma}$$

$$f_{\sigma}(\sigma) = f(\log \sigma) \frac{1}{\sigma}$$

$$a, \sigma \mid y_1, y_2$$

$$f(a, \sigma) \propto \exp\left(-\frac{(y_1 - a)^2 + (y_2 - a)^2}{2\sigma^2}\right) \frac{1}{\sigma^3}$$

$$a, \sigma \mid y_1, y_2$$

$$f_a(a) = \int_0^{\infty} f_{a, \sigma \mid y_1, y_2}(a, \sigma) d\sigma$$

$$\propto \int_0^{\infty} \sigma^{-3} \exp\left(-\frac{(y_1 - a)^2 + (y_2 - a)^2}{2\sigma^2}\right) d\sigma$$

$$\frac{\sigma}{\sqrt{(y_1 - a)^2 + (y_2 - a)^2}} = t$$

$$= \int_0^{\infty} t^{-3} \left((y_1 - a)^2 + (y_2 - a)^2 \right)^{-\frac{3}{2}} \exp\left(-\frac{1}{2t}\right) dt$$

$$\left[(y_1 - a)^2 + (y_2 - a)^2 \right]^{1/2}$$

$$= \left(\frac{1}{(y_1 - a)^2 + (y_2 - a)^2} \right) \int_0^{\infty} t^{-3} \exp\left(-\frac{1}{2t}\right) dt$$

$$f(a) \propto \frac{1}{(y_1 - a)^2 + (y_2 - a)^2}$$

$a | y_1, y_2$

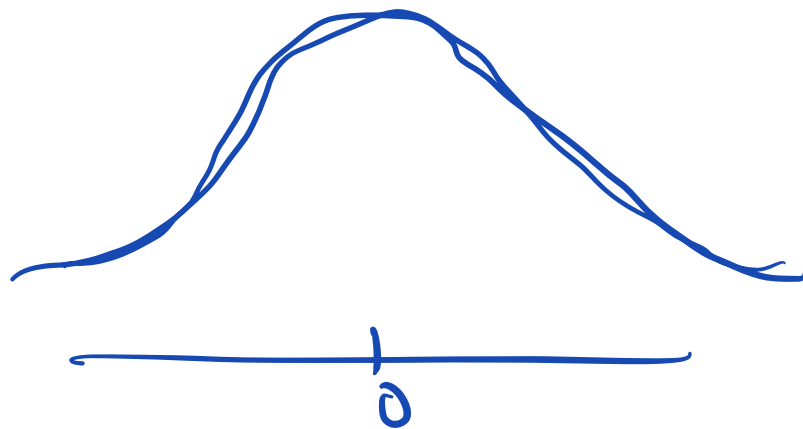
$$y_1 = 7, \quad y_2 = 10$$

$$\left(\frac{1}{(a - 7)^2 + (a - 10)^2} \right)$$

$$\frac{1}{(a - y_1)^2 + (a - y_2)^2}$$

$$\frac{1}{\pi} \frac{1}{1 + x^2} \rightarrow C(0, 1)$$

Standard



$C(\mu, \tau)$

$$\frac{1}{\pi} \frac{1}{1 + \left(\frac{x - \mu}{\tau}\right)^2} \times \frac{1}{\tau}$$

$$\propto \frac{1}{\tau^2 + (x - \mu)^2} \left\{ \frac{1}{(x - y_1)^2 + (x - y_2)^2} \right\}$$

$$\begin{aligned} (x - y_1)^2 + (x - y_2)^2 \\ = 2 \left(x - \frac{y_1 + y_2}{2} \right)^2 + \frac{(y_1 - y_2)^2}{2} \end{aligned}$$

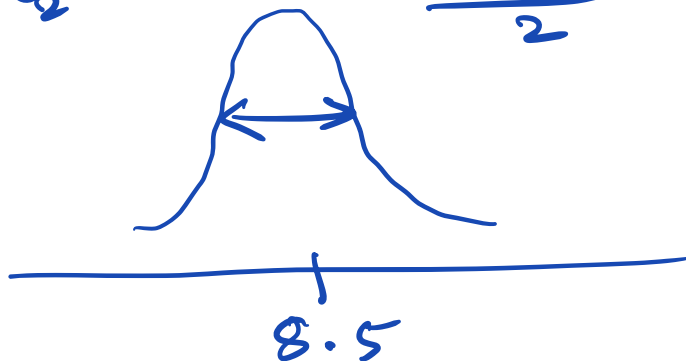
$$\frac{(y_1 - y_2)^2}{2} + 2 \left(x - \frac{y_1 + y_2}{2} \right)^2$$

Ans: $a \mid y_1, y_2 \sim C \left(\frac{y_1 + y_2}{2}, \frac{|y_1 - y_2|}{2} \right)$

$$a + b_1 = 7$$

$$a + b_2 = 10$$

$$\frac{|y_1 - y_2|}{2}$$



Question 4: Suppose there is an unknown physical quantity θ . A scientist makes $n = 6$ measurements on θ & they obtain $y_1 = 26.6$, $y_2 = 38.5$, $y_3 = 34.4$, $y_4 = 34$, $y_5 = 31$, $y_6 = 23.6$. On the basis of these measurements, what can you say about θ ?

Ans: $y_i = \theta + \varepsilon_i$

$$\boxed{y_i = \theta + \varepsilon_i}, \quad i = 1, \dots, n$$

$$\theta \sim \text{Unif}(-C, C), \quad \varepsilon_1, \dots, \varepsilon_n \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

$\log \sigma \sim \text{Unif}(-C, C)$

Goal: $\theta \mid y_1 = 26.6, \dots, y_6 = 23.6$

You get the t -distribution.

$$f(\theta, \sigma) | y_1, \dots, y_n$$

$$\propto f(y_1, \dots, y_n | \theta, \sigma) \boxed{f(\theta, \sigma)}$$

$$= \underbrace{\left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \theta)^2}{2\sigma^2}\right) \right]}_{\text{likelihood}} \frac{I\{-C < \theta < C\}}{2C}$$

$$\frac{I\{-C < \ln \sigma < C\}}{2C\sigma}$$

$$\propto \sigma^{-n-1} \exp\left[-\frac{\sum_{i=1}^n (y_i - \theta)^2}{2\sigma^2}\right]$$

$$\boxed{S(\theta) = \sum_{i=1}^n (y_i - \theta)^2} \rightarrow \text{Squares}$$

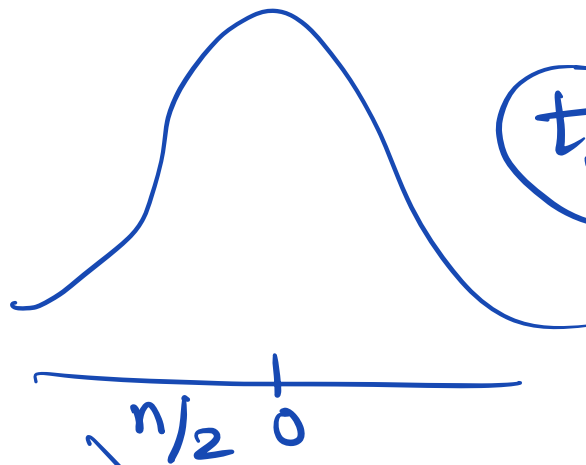
$$\sigma^{-n-1} \exp\left(-\frac{S(\theta)}{2\sigma^2}\right)$$

$$f_{\theta | y_1, \dots, y_n}(\theta) \propto \int_0^{\infty} \sigma^{-n-1} \exp\left(-\frac{S(\theta)}{2\sigma^2}\right) d\sigma$$

$$\propto \left[\frac{1}{S(\theta)} \right]^{n/2}$$

$$\left[\frac{1}{(\theta - y_1)^2 + \dots + (\theta - y_n)^2} \right]^{n/2}$$

$$\frac{y_1 + \dots + y_n}{n}$$



$$t_{n-1}$$

$$f(\theta) \propto$$

$\theta \mid y_1, \dots, y_n$

$$\left(\frac{1}{S(\theta)} \right)^{n/2}$$

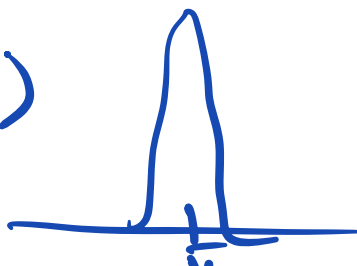
$$\hat{\theta} = \frac{y_1 + \dots + y_n}{n}$$

minimizes $S(\theta)$

$$\sum_{i=1}^n (y_i - \theta)^2$$

$$\propto \left(\frac{S(\hat{\theta})}{S(\theta)} \right)^{n/2}$$

$$S(\theta) = (1.1) S(\hat{\theta})$$



Question 5: y_t time series
 $t = 1 \dots n$

$$y_i = \beta_0 + \beta_1 i + \varepsilon_i^L \quad \begin{matrix} n+2 \\ n+1000 \end{matrix}$$

linear regression with
time & covariate: $x_i = i$ ✓

Lagged regression: $x_i = y_{i-1}$

Assume: $\varepsilon_1, \dots, \varepsilon_n \text{ iid } N(0, \sigma^2)$ ✓

$$(\beta_0, \beta_1) \sim \text{Unif}(-C, C) \quad \checkmark$$

$$\log \sigma \sim \text{Unif}(-C, C)$$

$$\beta_0, \beta_1 \mid y_1, \dots, y_n \quad \begin{matrix} \text{sm. OLS} \\ (y, x) \end{matrix}$$

$$f(\beta_0, \beta_1, \sigma) \mid y_1, \dots, y_n$$

$$\propto f(y_1, \dots, y_n \mid \beta_0, \beta_1, \sigma) f(\beta_0, \beta_1, \sigma)$$

$$= \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 i)^2}{2\sigma^2}\right) \right] \frac{I(-C < \beta_0 < C)}{2C} \frac{I(-C < \beta_1 < C)}{2C}$$

$$\frac{I(-C < \ln \sigma < C)}{2C\sigma}$$

$$\propto \sigma^{-n-1} \exp \left[-\frac{S(\beta_0, \beta_1)}{2\sigma^2} \right]$$

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 i)^2$$

$$f(\beta_0, \beta_1) \propto \left(\frac{1}{S(\beta_0, \beta_1)} \right)^{n/2} \rightarrow \text{Multivariate } t\text{-density}$$